4.1 VECTOR SPACES & SUBSPACES

Many concepts concerning vectors in R^n can be extended to other mathematical systems.

We can think of a vector space in general, as a collection of objects that behave as vectors do in R^n. The objects of such a set are called vectors.

A vector space is a nonempty set V of objects, called vectors, on which are defined two operations, called addition and multiplication by scalars (real numbers), subject to the ten axioms below. The axioms must hold for all **u, v** and **w** in V and for all scalars c and d.

1. $\mathbf{u} + \mathbf{v}$ is in V.
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
4. There is a vector (called the zero vector) 0 in V such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
5. For each $\mathbf{u}$ in V, there is vector $-\mathbf{u}$ in V satisfying $\mathbf{u} + (-\mathbf{u}) = 0$.
6. $c\mathbf{u}$ is in V.
7. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$.
8. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$.
9. $(cd)\mathbf{u} = c(d\mathbf{u})$.
10. $1\mathbf{u} = \mathbf{u}$.

Common Vector Spaces

R^1—the set of all real numbers

R^2—the set of ordered pairs of real numbers (the xy-plane)

R^3—three-dimensional Euclidean space

R^n—the set of all column (or row vectors with n components where the components are real numbers

Vector spaces where the scalars are complex numbers are called complex vector spaces. If the scalars are real, they are called real vector spaces.

The simplest of all vector spaces is the zero vector space.

Vector Space Examples

Ex 1: Let $M_{2x2} = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \text{ are real} \right\}$ (In this context, what is the **0** vector?)

Ex 2: Let $n \geq 0$ be an integer and let $\mathbf{P}_n$ = the set of all polynomials of degree at most $n \geq 0$.

Members of $\mathbf{P}_n$ have the form $\mathbf{p}(t) = a_0 + a_1 t + a_2 t^2 + \cdots + a_n t^n$ where $a_0, a_1, \ldots, a_n$ are real numbers and t is a real variable. The set $\mathbf{P}_n$ is a vector space. To prove it is a vector space, we must prove that all 10 axioms hold. We will just verify 3 out of the 10 axioms here.

Let $\mathbf{p}(t) = a_0 + a_1 t + a_2 t^2 + \cdots + a_n t^n$ and $\mathbf{q}(t) = b_0 + b_1 t + b_2 t^2 + \cdots + b_n t^n$. Let c be a scalar.

Axiom 1 : The polynomial $\mathbf{p} + \mathbf{q}$ is defined as follows: $(\mathbf{p} + \mathbf{q})(t) = \mathbf{p}(t) + \mathbf{q}(t)$. Therefore, $(\mathbf{p} + \mathbf{q})(t) = \mathbf{p}(t) + \mathbf{q}(t) =$ (_____) + (_____)t + . . . + (_____)t^n which is also a _____ of degree at most _____. So $\mathbf{p} + \mathbf{q}$ is in $\mathbf{P}_n$

Axiom 4: $\mathbf{0} = 0 + 0t + \cdots + 0t^n$ (zero vector in $\mathbf{P}_n$)

$(\mathbf{p} + 0)(t) = \mathbf{p}(t) + \mathbf{0} = (a_0 + 0) + (a_1 + 0)t + \cdots + (a_n + 0)t^n = a_0 + a_1 t + \cdots + a_n t^n = \mathbf{p}(t)$ and so $\mathbf{p} + \mathbf{0} = \mathbf{p}$

Axiom 6: $(c\mathbf{p})(t) = c\mathbf{p}(t) =$ (_____) + (_____)$t + \cdots +$ (_____)t^n which is in $\mathbf{P}_n$

The other 7 axioms also hold, so $\mathbf{P}_n$ is a vector space.

There are important vector spaces "inside" the standard spaces R^n. For example, consider R^3 and choose any plane that passes through the origin—that plane is a vector space in its own right

- The zero vector is part of the set
- If we multiply a vector in the plane by a scalar, we get a vector that lies in the same plane
- If we add any 2 vectors in the plane we get a vector in the plane.

This plane illustrates on of the most fundamental ideas in linear algebra—it is a subspace of the original space R^3

Subspaces: Vector spaces may be formed from subsets of other vectors spaces. These are called subspaces.
A subspace of a vector space V is a subset H of V that has three properties:

a. The zero vector of V is in H.
b. For each $\mathbf{u}$ and $\mathbf{v}$ are in H, $\mathbf{u} + \mathbf{v}$ is in H. (In this case we say H is closed under vector addition.)
c. For each $\mathbf{u}$ in H and each scalar c, $c\mathbf{u}$ is in H. (In this case we say H is closed under scalar multiplication.)

If the subset H satisfies these three properties, then H itself is a vector space.
The smallest subspace is the zero vector.
The empty set is not a subspace.

Ex 3: Let $H = \left\{ \begin{bmatrix} a \\ 0 \\ b \end{bmatrix} : a, b \text{ are real} \right\}$. Show that H is a subspace of R^3.

Ex 4: Is $H = \left\{ \begin{bmatrix} x \\ x+1 \end{bmatrix} : x \text{ is a real number} \right\}$ a subspace of R^2?

A Shortcut for Determining Subspaces

Theorem 1: If $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are in a vector space V, then Span$\{ \mathbf{v}_1, \ldots, \mathbf{v}_p \}$ is a subspace of V.

Proof: In order to verify this, check properties a, b and c of definition of a subspace.

a. $\mathbf{0}$ is in Span$\{ \mathbf{v}_1, \ldots, \mathbf{v}_p \}$ since $\mathbf{0} =$ _____$\mathbf{v}_1$ + _____$\mathbf{v}_2$ + ⋯ + _____ $\mathbf{v}_p$

b. To show that Span$\{ \mathbf{v}_1, \ldots, \mathbf{v}_p \}$ is closed under vector addition, we choose two arbitrary vectors in Span$\{ \mathbf{v}_1, \ldots, \mathbf{v}_p \}$:
$\mathbf{u} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n$ and $\mathbf{v} = b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \cdots + b_n\mathbf{v}_n$ Then

$\mathbf{u} + \mathbf{v} =$ (_____) + (_____) =

(_____$\mathbf{v}_1$ + _____$\mathbf{v}_1$) + (_____$\mathbf{v}_2$ + _____$\mathbf{v}_2$) + ⋯ + (_____$\mathbf{v}_p$ + _____$\mathbf{v}_p$)

So $\mathbf{u}+\mathbf{v}$ is in Span$\{ \mathbf{v}_1, \ldots, \mathbf{v}_p \}$

c. To show that Span$\{ \mathbf{v}_1, \ldots, \mathbf{v}_p \}$ closed under scalar multiplication, choose an arbitrary number c and an arbitrary vector in Span$\{ \mathbf{v}_1, \ldots, \mathbf{v}_p \}$, $\mathbf{v} = b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \cdots + b_n\mathbf{v}_n$

Then $c\mathbf{v} = c(b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \cdots + b_n\mathbf{v}_n) =$ _____$\mathbf{v}_1$ + _____$\mathbf{v}_2$ + … + _____$\mathbf{v}_p$
So $c\mathbf{v}$ is in Span$\{ \mathbf{v}_1, \ldots, \mathbf{v}_p \}$,

Since properties a, b and c hold, Span$\{ \mathbf{v}_1, \ldots, \mathbf{v}_p \}$, is a subspace of V.

Recap

1.To show that H is a subspace of a vector space, use Theorem 1.
2.To show that a set is not a subspace of a vector space, provide a specific example showing that at least one of the axioms a, b or c (from the definition of a subspace) is violated.

Ex 5: Is $V=\{(a+2b, 2a-3b)$: a and b are real$\}$ a subspace of R^2? Why or why not?

Ex 6: Is $H=\left\{\begin{bmatrix} a+2b \\ a+1 \\ a \end{bmatrix} : a \text{ and } b \text{ are real}\right\}$ a subspace of R^3? Why or why not?

Ex 7: Is the set H of all matrices of the form $\begin{bmatrix} 2a & b \\ 3a+b & 3b \end{bmatrix}$ a subspace of M_{2x2} Explain.

4.2 Null Spaces, Column Spaces & Linear Transformations

The **null space** of an *m x n* matrix A, written as **Nul A**, is the set of all solutions to the homogeneous equation A**x** = **0**.

Nul A = {**x**: **x** is in R^n and A**x** = **0**}

Theorem 2

The null space of an *m x n* matrix A is a subspace of R^n. Equivalently, the set of all solutions to a system A**x** = **0** of *m* homogeneous linear equations in *n* unknowns is a subspace of R^n.

Proof: Nul A is a subset of R^n since A has *n* columns. Must verify properties a, b, and c of the definition of a subspace.

Solving Ax = 0 yields an explicit description of Nul A.

Ex 1: Find an explicit description of Nul A where A = $\begin{bmatrix} 3 & 6 & 6 & 3 & 9 \\ 6 & 12 & 13 & 0 & 3 \end{bmatrix}$

Observations:

1. Spanning set of Nul A, found using the method in the last example, is automatically linear independent:

2. If $NulA \neq \{0\}$, then the number of vectors in the spanning set for Nul A equals the number of free variables in A**x** = **0**.

The **column space** of an *m x n* matrix A (**Col A**) is the set of all linear combinations of the columns of A.

If A = $[a_1 \cdots a_n]$, then Col A = Span$\{a_1, \ldots, a_n\}$

Theorem 3: The column space of an *m x n* matrix A is a subspace of R^m.

Why? (Theorem 1, page 194)

Recall that if A**x** = b, then b is a linear combination of A. Therefore,Col A = {b: b = A**x** for some **x** in R^n}

Ex 2: Find a matrix A such that W = Col A where $W = \left\{ \begin{bmatrix} x-2y \\ 3y \\ x+y \end{bmatrix} : x, y \text{ in } \mathbb{R} \right\}$

The column space of an *m x n* matrix A is all of R^m if and only if the equation A**x** = b has a solution for each b in R^m. Why?

Ex 3: Is w in Nul A? $w = \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix}, A = \begin{bmatrix} 5 & 21 & 19 \\ 13 & 23 & 2 \\ 8 & 14 & 1 \end{bmatrix}$

Ex 4: Is w in Col A?

The Contrast Between Nul A and Col A

Ex 5: Let $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \\ 0 & 0 & 1 \end{bmatrix}$

(a) The column space of A is a subspace of R^k where k = _____.

(b) The null space of A is a subspace of R^k where k = _____.

(c) Find a nonzero vector in Col A. (There are infinitely many possibilities.)

$$-\begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix} + -\begin{bmatrix} 2 \\ 4 \\ 6 \\ 0 \end{bmatrix} + -\begin{bmatrix} 3 \\ 7 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} \quad \\ \quad \\ \quad \\ \quad \end{bmatrix}$$

(d) Find a nonzero vector in Nul A. Solve A**x** = 0 and pick one solution.

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 7 & 0 \\ 3 & 6 & 10 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1 = -2x_2$, x_2 is free, $x_3 = 0$

Let x_2 = _____ and then **x** = $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \quad \\ \quad \\ \quad \end{bmatrix}$

Contrast Between Nul A and Col A for an m x n matrix A (see page 204)

Nul A	**Col A**
1. Subspace of R^n	1. Subspace of R^m
2. Implicit	2. Explicit
3. Takes time to find vectors in Nul A	3. Easy to find vectors in Col A
4. No obvious relationship between Nul A and the entries in A	4. Obvious relationship between Col A and entries in A
5. A**v** = 0, v in Nul A	5. A**x** = v consistent, v in Col A
6. Easy to tell if v is in Nul A, compute Av	6. May take time to tell if v is in Col A, Do [A v]
7. Nul A = {0} iff A**x** = 0 has only trivial solution	7. Col A = R^m iff A**x** = b has a solution for every b in R^m
8. Nul A = {0} iff linear transformation $x \rightarrow Ax$ is one-to-one	8. Col A = R^m iff linear transformation $x \rightarrow Ax$ maps R^n onto R^m

Review:

A subspace of a vector space V is a subset H of V that has three properties:

a. The zero vector of V is in H.
b. For each **u** and **v** in H, **u** + **v** is in H.
(closed under addition)
c. For each **u** in H and each scalar c, c**u** is in H.
(closed under scalar multiplication)

If the subset H satisfies these three properties, then H itself is a vector space.

Theorem 1, 2, and 3 (Sections 4.1 & 4.2)
If $v_1, \bullet\ \bullet\ \bullet, v_p$ are in a vector space V, then Span $\{v_1, \bullet\ \bullet\ \bullet, v_p\}$ is a subspace of V.
The null space of an *m x n* matrix A is a subspace of R^n.
The column space of an *m x n* matrix A is a subspace of R^m.

Ex 6: Determine whether each of the following sets is a vector space or provide a counterexample.

(a) $H = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x - y = 4 \right\}$

(b) $V = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x - y = 0 \text{ and } y + z = 0 \right\}$

(c) $S = \left\{ \begin{bmatrix} x + y \\ 2x - 3y \\ 3y \end{bmatrix} : x, y, z \text{ are real} \right\}$

Kernel and Range of a Linear Transformation

A linear transformation T from a vector space V into a vector space W is a rule that assigns to each **x** in V a unique vector T(**x**) in W, such that

i. T(**u** + **v**) = T(**u**) + T(**v**) for all **u**,**v** in V
ii. T(c**u**) = cT(**u**) for all **u** in V and all scalars c.

The **kernel** (or **null space**) of T is the set of all vectors **u** in V such that T(**u**) = 0.

The range of T is the set of all vectors in W of the form T(**u**) where **u** is in V.

So if T(**x**) = A**x**, Col A = range of T

Ex 7: $L : \mathbb{R}^4 \to \mathbb{R}^2$ $\quad \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} x + y \\ z + w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ What form do the vectors in **ker L** look like?

Ex 8: $L:\mathbb{R}^3 \to \mathbb{R}^2$ mapping $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ to $\begin{bmatrix} x+2y+z \\ 2x-z \end{bmatrix}$.

Is (0, 0, 2) in **ker L**?

Is (2, -3, 4) in **ker L**?

Ex 9: Let A = $\begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix}$ and w = $\begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$

Is w in **Nul A**?

Is w in **Col A**?

4.3 Linearly Independent Sets

A set of vectors $\{ \mathbf{v}_1, \ldots, \mathbf{v}_p \}$ in a vector space V is said to be linearly independent if the vector equation $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \ldots + c_p \mathbf{v}_p = \mathbf{0}$ has only the trivial solution $c_1=0,\ldots,c_p=0$.

The set $\{ \mathbf{v}_1, \ldots, \mathbf{v}_p \}$ is said to be linearly dependent if there exists weights $c_1=0,\ldots,c_p$, not all 0, such that $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \ldots + c_p \mathbf{v}_p = \mathbf{0}$

The following results from Section 1.7 are still true for more general vectors spaces.

A set containing the zero vector is linearly dependent.

A set of two vectors is linearly dependent if and only if one is a multiple of the other.

Ex 1: Determine if $\left\{ \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 3 & 2 \\ 3 & 0 \end{bmatrix} \right\}$ is a linearly dependent or independent set.

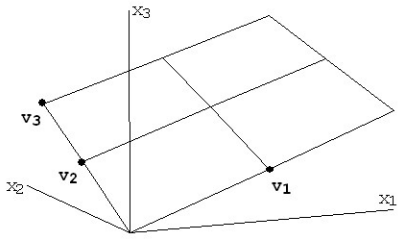
Ex 2: Determine if $\left\{ \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 3 & 6 \\ 9 & 10 \end{bmatrix} \right\}$ is a linearly dependent or independent set.

Theorem 4: An indexed set $\{ \mathbf{v}_1, \ldots, \mathbf{v}_p \}$ of two or more vectors, with $\mathbf{v}_j \neq 0$, is linearly dependent if and only if some vector $\mathbf{v}_j$ ($j > 1$) is a linear combination of the preceding vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$

Ex 3: Let $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ be a set of vectors in $\mathbf{P}_2$ where $\mathbf{p}_1(t) = t$, $\mathbf{p}_2(t) = t^2$, and $\mathbf{p}_3(t) = 4t + 2t^2$. Is this a linearly dependent set?

A Basis Set Let H be the plane illustrated below. Which of the following are valid descriptions of H?

(a) H=Span$\{\mathbf{v}_1, \mathbf{v}_2\}$
(b) H=Span$\{\mathbf{v}_1, \mathbf{v}_3\}$
(c) H=Span$\{\mathbf{v}_2, \mathbf{v}_3\}$
(d) H=Span$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$



A basis set is an "efficient" spanning set containing no unnecessary vectors. In this case, we would consider the linearly independent sets $\{\mathbf{v}_1, \mathbf{v}_2\}$ and $\{\mathbf{v}_1, \mathbf{v}_3\}$ to both be examples of basis sets or bases (plural for basis) for H.

Definition: Let H be a subspace of a vector space V. An indexed set of vectors $\beta = \{\mathbf{b}_1, \ldots, \mathbf{b}_p\}$ in V is a basis for H if

(i) β is a linearly independent set, and

(ii) H= Span$\{ \mathbf{b}_1, \ldots, \mathbf{b}_p \}$.

Ex 4: Let $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Show that $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is a basis for R^3. The set $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is called a standard basis for R^3.
(Hint: Use the IMT)

Ex 5: Let $S = \{1, t, t^2, \ldots, t^n\}$. Show that S is a basis for $\mathbf{P}_n$.

Ex 6: Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$. Is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ a basis for R^3?

Ex 7: Explain why each of the following sets is not a basis for $\mathbb{R}^3$.

(a) $\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}4\\5\\7\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}1\\-3\\7\end{bmatrix}\right\}$ (b) $\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}4\\5\\6\end{bmatrix}\right\}$

Bases for Nul A

Ex 8 Find a basis for Nul A where $A = \begin{bmatrix}3 & 6 & 6 & 3 & 9\\ 6 & 12 & 13 & 0 & 3\end{bmatrix}$. Hint: Row reduce $[A\ \mathbf{0}]$

In this example, the set of three vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a spanning set for Nul A. In the last section we observed that this set is linearly independent. Therefore $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a basis for Nul A. The technique used here always provides a linearly independent set.

The Spanning Set Theorem: Informally, this theorem says a basis can be constructed from a spanning set of vectors by discarding vectors which are linear combinations of preceding vectors in the indexed set.

Ex 9: Find a basis for the set of vectors
$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ if $\mathbf{v}_1 = \begin{bmatrix}-1\\0\end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix}0\\-1\end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix}-2\\-3\end{bmatrix}$

Theorem 5 (The Spanning Set Theorem): Let $S=\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ be a set in V and let $H=$ Span$\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$
a. If one of the vectors in S, say $\mathbf{v}_k$, is a linear combination of the remaining vectors in S, then the set formed from S by removing $\mathbf{v}_k$ still spans H.
b. If $H \neq \{\mathbf{0}\}$, some subset of S is a basis for H.

Bases for Col A

Ex 10: Find a basis for Col A, where $A=[\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3\ \mathbf{a}_4] = \begin{bmatrix}1 & 2 & 0 & 4\\ 2 & 4 & -1 & 3\\ 3 & 6 & 2 & 22\\ 4 & 8 & 0 & 16\end{bmatrix}$

Elementary row operations on a matrix do not affect the linear dependence relations among the columns of the matrix.

Theorem 6: The pivot columns of a matrix A form a basis for Col A.

Ex 11: Let $\mathbf{v}_1=\begin{bmatrix}1\\2\\-3\end{bmatrix}$, $\mathbf{v}_2=\begin{bmatrix}-2\\-4\\6\end{bmatrix}$, $\mathbf{v}_3=\begin{bmatrix}3\\6\\9\end{bmatrix}$. Find a basis for Span$\{\mathbf{v}_1,\mathbf{v}_2,\mathbf{v}_3\}$.

Review:

1. To find a basis for Nul A, use elementary row operations to transform $[A\ \ \mathbf{0}]$ to an equivalent reduced row echelon form $[B\ \ \mathbf{0}]$. Use the reduced row echelon form to find parametric form of the general solution to $A\mathbf{x}=0$. The vectors found in this parametric form of the general solution form a basis for Nul A.
2. A basis for Col A is formed from the pivot columns of A. Warning: Use the pivot columns of A, not the pivot columns of B, where B is in reduced echelon form and is row equivalent to A.

4.4 Coordinate Systems

In general, people are more comfortable working with the vector space R^n and its subspaces than with other types of vectors spaces and subspaces. The goal here is to impose coordinate systems on vector spaces, even if they are not in R^n.

Theorem 7 The Unique Representation Theorem: Let $\beta=\{\mathbf{b}_1,\ldots,\mathbf{b}_n\}$ be a basis for a vector space V. Then for each $\mathbf{x}$ in V, there exists a unique set of scalars $c_1,\ldots,c_n$ such that $\mathbf{x}=c_1\mathbf{b}_1+\cdots+c_n\mathbf{b}_n$

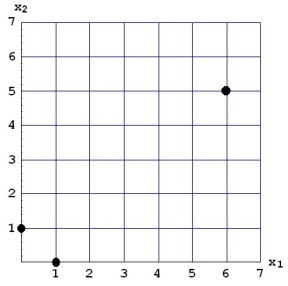
Definition: Suppose $\beta=\{\mathbf{b}_1,\ldots,\mathbf{b}_n\}$ is a basis for a vector space V and $\mathbf{x}$ is in V. The coordinates of $\mathbf{x}$ relative to the basis β (or the β-coordinates of $\mathbf{x}$) are the weights $c_1,\ldots,c_n$ such that $\mathbf{x}=c_1\mathbf{b}_1+\cdots+c_n\mathbf{b}_n$

In this case, the vector in R^n $[\mathbf{x}]_\beta=\begin{bmatrix}c_1\\\vdots\\c_n\end{bmatrix}$ is called the coordinate vector of $\mathbf{x}$ (relative to β), or the β- coordinate vector of $\mathbf{x}$

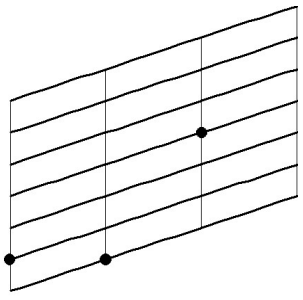
Ex 1: Let $\beta=\{\mathbf{b}_1,\mathbf{b}_2\}$ where $\mathbf{b}_1=\begin{bmatrix}3\\1\end{bmatrix}$ and $\mathbf{b}_2=\begin{bmatrix}0\\1\end{bmatrix}$ and let $E=\{\mathbf{e}_1,\mathbf{e}_2\}$ where $\mathbf{e}_1=\begin{bmatrix}1\\0\end{bmatrix}$ and $\mathbf{e}_2=\begin{bmatrix}0\\1\end{bmatrix}$. Express $[\mathbf{x}]_\beta=\begin{bmatrix}2\\3\end{bmatrix}$ in terms of the $E=\{\mathbf{e}_1,\mathbf{e}_2\}$; i.e., find $[\mathbf{x}]_E$.

Geometrically, we are changing from using basis vectors <3,1> and <0,1> to standard basis vectors <1, 0> and <0, 1>.

Standard graph paper



β- graph paper



From the last example, $\begin{bmatrix}6\\5\end{bmatrix}=\begin{bmatrix}3&0\\1&1\end{bmatrix}\begin{bmatrix}2\\3\end{bmatrix}$

For a basis $\beta = \{\mathbf{b}_1,\ldots,\mathbf{b}_n\}$ let $P_\beta = [\mathbf{b}_1,\ldots,\mathbf{b}_n]$ and $[\mathbf{x}]_\beta = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$ Then $\mathbf{x} = P_\beta\,[\mathbf{x}]_\beta$

We call P_β the change-of-coordinates matrix from β to the standard basis in R^n. Then $[\mathbf{x}]_\beta = P_\beta^{-1}\,\mathbf{x}$

and therefore P_β^{-1} is a change-of-coordinates matrix from the standard basis in R^n to the basis β.

Ex 2: Let $\beta=\{\mathbf{b}_1,\mathbf{b}_2\}$ where $\mathbf{b}_1=\begin{bmatrix}3\\1\end{bmatrix}$, $\mathbf{b}_2=\begin{bmatrix}0\\1\end{bmatrix}$ and $\mathbf{x}=\begin{bmatrix}6\\8\end{bmatrix}$. Find the change-of-coordinates matrix P_β from β to the standard basis in R^2 and change-of-coordinates matrix P_β^{-1} from the standard basis in R^2 to β.

Ex 3: Find the vector $\mathbf{x}$ determined by the given coordinate vector $[\mathbf{x}]_\beta$ and the given basis β.

$\beta = \left\{\begin{bmatrix}4\\5\end{bmatrix},\begin{bmatrix}6\\7\end{bmatrix}\right\}$, $[\mathbf{x}]_\beta = \begin{bmatrix}8\\-5\end{bmatrix}$

Ex 4: Find the coordinate vector $[\mathbf{x}]_\beta$ of $\mathbf{x}$ relative to the given basis $\beta = \{\mathbf{b}_1,\ldots,\mathbf{b}_n\}$ $\mathbf{b}_1 = \begin{bmatrix}1\\-2\end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix}5\\-6\end{bmatrix}$, $\mathbf{x} = \begin{bmatrix}4\\0\end{bmatrix}$

Coordinate mappings allow us to introduce coordinate systems for unfamiliar vector spaces.

Standard basis for $\mathbf{P}_2$: $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\} = \{1, t, t^2\}$

Polynomials in $\mathbf{P}_2$ behave like vectors in R^3.

Since $a + bt + ct^2 = a\mathbf{p}_1 + b\mathbf{p}_2 + c\mathbf{p}_3$, $[a + bt + ct^2]_\beta = \begin{bmatrix}a\\b\\c\end{bmatrix}$

We say that the vector space R^3 is isomorphic to $\mathbf{P}_2$.

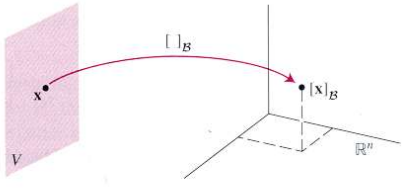


FIGURE 5 The coordinate mapping from V onto $\mathbb{R}^n$.

Parallel Worlds of R^3 and $\mathbf{P}_2$

Vector Space R^3	Vector Space $\mathbf{P}_2$
Vector Form: $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$	Vector Form: $a + bt + ct^2$
Vector Addition Example $\begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$	Vector Addition Example $(-1 + 2t - 3t^2) + (2 + 3t + 5t^2) = 1 + 5t + 2t^2$

Informally, we say that vector space V is isomorphic to W if every vector space calculation in V is accurately reproduced in W, and vice versa.

Assume β is a basis set for vector space V. Exercise 25 (page 223) shows that a set $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ in V is linearly independent if and only if $\{[\mathbf{u}_1]_\beta, [\mathbf{u}_2]_\beta, \ldots, [\mathbf{u}_p]_\beta\}$ is linearly independent in R^n.

Ex 5: Use coordinate vectors to determine if $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ is a linearly independent set, where $\mathbf{p}_1 = 1 - t$, $\mathbf{p}_2 = 2 - t + t^2$, and $\mathbf{p}_3 = 2t + 3t^2$.

Ex 6: Determine whether the set of polynomials forms a basis for $\mathbf{P}_3$. $\{5 - 3t + 4t^2 + 2t^3, 9 + t + 8t^2 - 6t^3, 6 - 2t + 5t^2, t^3\}$

Coordinate vectors also allow us to associate vector spaces with subspaces of other vectors spaces.

Ex 7: Let $\beta = \{\mathbf{b}_1, \mathbf{b}_2\}$ where $\mathbf{b}_1 = \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$ and let $H = \text{span}\{\mathbf{b}_1, \mathbf{b}_2\}$. Find $[\mathbf{x}]_\beta$, if $\mathbf{x} = \begin{bmatrix} 9 \\ 13 \\ 15 \end{bmatrix}$

4.5 The Dimension of a Vector Space

Theorem 9: If a vector space V has a basis $\beta = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$, then any set in V containing more than n vectors must be linearly dependent.

Proof: Suppose $\{u_1, \ldots, u_p\}$ is a set of vectors in V where p > n. Then the coordinate vectors $\{[u_1]_\beta, \cdots, [u_p]_\beta\}$ are in R^n. Since $p > n$, $\{[u_1]_\beta, \cdots, [u_p]_\beta\}$ are linearly dependent and therefore $\{u_1, \ldots, u_p\}$ are linearly dependent.

Theorem 10: If a vector space V has a basis of n vectors, then every basis of V must consist of n vectors.

Proof: Suppose β_1 is a basis for V consisting of exactly n vectors. Now suppose β_2 is any other basis for V. By definition of a basis, we know that β_1 and β_2 are both linearly independent sets.

By Theorem 9, if β_1 has more vectors than β_2, then _____ is a linearly dependent set (which cannot be the case).

Again by Theorem 9, if β_2 has more vectors than β_1, then _____ is a linearly dependent set (which cannot be the case).

Therefore β_2 has exactly n vectors also.

Definition – If V is spanned by a finite set, then V is said to be finite-dimensional, and the dimension of V, written as dim V, is the number of vectors in a basis for V. The dimension of the zero vector space $\{0\}$ is defined to be 0. If V is not spanned by a finite set, then V is said to be infinite-dimensional.

Ex 1: The standard basis for $\mathbf{P}_3$ is {_____}. So dim P_3 = _____. In general, dim $P_n = n + 1$.

The standard basis for R^n is $\{e_1, \ldots, e_n\}$ where $e_1, \ldots, e_n$ are the columns of I_n. So, for example, dim $R^3 = 3$.

Ex 2: Find a basis and the dimension of the subspace $W = \left\{ \begin{bmatrix} a+b+2c \\ 2a+2b+4c+d \\ b+c+d \\ 3a+3c+d \end{bmatrix} : a, b, c, d \text{ are real} \right\}$

Dimensions of subspaces of R^3

0-dimensional subspace contains only the zero vector $\left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$

1-dimensional subspaces. Span {**v**} where $v \neq 0$ is in R^3. These subspaces are _____ through the origin.

2-dimensional subspaces. Span {**u**, **v**} where **u** and **v** are in R^3 and are not multiples of each other. These subspaces are

_____ through the origin.

3-dimensional subspaces. Span {**u, v, w**} where u, v, w are linearly independent vectors in R^3. This subspace is R^3 itself because the columns of A = [**u v w**] span R^3 according to the IMT.

Theorem 11: Let H be a subspace of a finite-dimensional vector space V. Any linearly independent set in H can be expanded, if necessary, to a basis for H. Also, H is finite-dimensional and dim $H \leq$ dim V.

Let $H = span\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}1\\1\\0\end{bmatrix}\right\}$. Then H is a subspace of R^3 and dim $H <$ dim R^3. We could expand the spanning set $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}1\\1\\0\end{bmatrix}\right\}$ to $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}1\\1\\0\end{bmatrix},\begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$ to form a basis for R^3

Theorem 12--The Basis Theorem: Let V be a p-dimensional vector space, $p \geq 1$. Any linearly independent set of exactly p vectors in V is automatically a basis for V. Any set of exactly p vectors that spans V is automatically a basis for V.

Ex 3: Show that $\{t, 1-t, 1+t-t^2\}$ is a basis for $\mathbf{P}_2$.

Dimensions of Col A and Nul A: Recall our techniques to find basis sets for column spaces and null spaces.

Ex 4: Suppose $A = \begin{bmatrix}1 & 2 & 3 & 4\\2 & 4 & 7 & 8\end{bmatrix}$. Find dim Col A and dim Nul A.

dim Col A = number of pivot columns of A

dim Nul A = number of free variables of A

4.6 Rank

The set of all linear combinations of the row vectors of a matrix A is called the row space of A and is denoted by Row A.

For example, let $A = \begin{bmatrix} -1 & 2 & 3 & 6 \\ 2 & -5 & -6 & -12 \\ 1 & -3 & -3 & -6 \end{bmatrix}$ and $\mathbf{r}_1 = (-1, 2, 3, 6)$, $\mathbf{r}_2 = (2, -5, -6, -12)$, $\mathbf{r}_3 = (1, -3, -3, -6)$

Row A = Span$\{ \mathbf{r}_1 , \mathbf{r}_2 , \mathbf{r}_3 \}$(a subspace of R^4)

While it is natural to express row vectors horizontally, they can also be written as column vectors if it is more convenient.

Therefore, Col A^T = Row A

When we use row operations to reduce matrix A to matrix B, we are taking linear combinations of the rows of A to come up with B. We could reverse this process and use row operations on B to get back to A. Because of this, the row space of A equals the row space of B.

Theorem 13: If two matrices A and B are row equivalent, then their row spaces are the same. If B is in echelon form, the nonzero rows of B form a basis for the row space of A as well as B.

Ex 1: The matrices $A = \begin{bmatrix} -1 & 2 & 3 & 6 \\ 2 & -5 & -6 & -12 \\ 1 & -3 & -3 & -6 \end{bmatrix}$ *and* $B = \begin{bmatrix} -1 & 2 & 3 & 6 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ are row equivalent. Find a basis for row space, column space and null space of A. Also state the dimension of each.

Note the following:

dim Col A = # of pivots of A = # of nonzero rows in B = dim Row A

dim Nul A = # of free variables = # of nonpivot columns of A

Definition: The rank of A is the dimension of the column space of A.

rank A = dim Col A = # of pivot columns of A = dim Row A

rank A	+	dim Nul A	=	n
# of pivot columns of A		# of nonpivot columns of A		# of columns of A

Theorem 14 The Rank Theorem: The dimensions of the column space and the row space of an m x n matrix A are equal. This common dimension, the rank of A, also equals the number of pivot positions in A and satisfies the equation rank A + dim Nul $A = n$

Since $Row\ A = Col\ A^T$, rank A = rank A^T .

Ex 2: Suppose that a 5 x 8 matrix A has rank 5. Find dim Nul A, dim Row A and rank A^T . Is Col $A = R^5$?

Ex 3: For a 9 x 12 matrix A, find the smallest possible value of dim Nul A.

Ex 4: A scientist solves a homogeneous system of 50 equations in 54 variables and finds that exactly 4 of the unknowns are free variables. Can the scientist be certain that any associated nonhomogeneous system (with the same coefficients) has a solution?

THE INVERTIBLE MATRIX THEOREM (continued)

Let A be a square n x n matrix. Then the following statements are equivalent:

m. The columns of A form a basis for R^n
n. Col $A = R^n$
o. dim Col $A = n$
p. rank $A = n$
q. Nul $A = \{\mathbf{0}\}$
r. dim Nul A = 0

Ex 5: If a 6x3 matrix A has rank 3, find dim Nul A, dim Row A and rank A^T.

Ex 6: Suppose a 5 x 6 matrix A has 4 pivot columns

a) What is dim Nul A?

b)Is Col A in R^4?

Ex 7: If the null space of a 7 x 6 matrix A is 5 dimensional, what is the dimension of the column space of A?

Ex 8: If the Null Space of a 5 x 6 matrix A is 4-dimensional, what is the dimension of the row space of A?

Ex 9: If A is a 4 x 3 matrix, what is the largest possible dimension of the row space of A?

Ex 10: If A is a 6 x 4 matrix, what is the smallest possible dimension of Nul A?

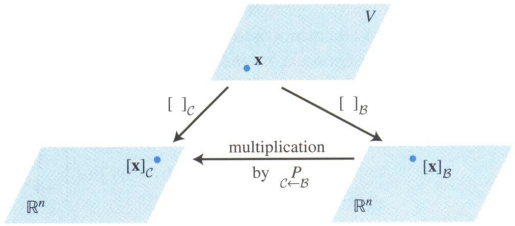
4.7 Change of Basis

When a basis β is chosen for an n-dimensional vector space V, the associated coordinate mapping onto R^n provides a coordinate system for V. Each $\mathbf{x}$ in V is identified uniquely by its β-coordinate vector $[\mathbf{x}]_\beta$.
Sometimes, a problem is described initially using a basis β, but the problem's solution is aided by changing β to a new basis C. Each vector is assigned a new C-coordinate vector.
In this section, we study how $[\mathbf{x}]_C$ and $[\mathbf{x}]_\beta$ are related for each $\mathbf{x}$ in V.

Theorem 15: Let $B = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ and $C = \{\mathbf{c}_1, \ldots, \mathbf{c}_n\}$ be bases of a vector space V. Then there is a unique n x n matrix $\underset{C\leftarrow B}{P}$ such that $[\mathbf{x}]_C = \underset{C\leftarrow B}{P}\,[\mathbf{x}]_B$.

The columns of $\underset{C\leftarrow B}{P}$ are the C-coordinate vectors of the vectors in the basis B. That is, $\underset{C\leftarrow B}{P} = [\,[\mathbf{b}_1]_C\ [\mathbf{b}_2]_C \ldots [\mathbf{b}_n]_C\,]$

The matrix $\underset{C\leftarrow B}{P}$ is called the change-of- coordinates matrix from B to C.



Multiplication by $\underset{C\leftarrow B}{P}$ converts B-coordinates into C-coordinates

Ex 1: Consider two bases $B = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $C = \{\mathbf{c}_1, \mathbf{c}_2\}$ for a vector space V such that $\mathbf{b}_1 = -\mathbf{c}_1 + 4\mathbf{c}_2$ and $\mathbf{b}_2 = 5\mathbf{c}_1 - 3\mathbf{c}_2$

(a) Find the change-of-coordinate matrix from B to C.

(b) Find $[\mathbf{x}]_C$ for $\mathbf{x} = 5\mathbf{b}_1 + 3\mathbf{b}_2$

The columns of $\underset{C\leftarrow B}{P}$ are linearly independent because they are the coordinate vectors of the linearly independent set B.
Since $\underset{C\leftarrow B}{P}$ is square, it must be invertible, by the IMT.

$\left(\underset{C\leftarrow B}{P}\right)^{-1}$ is the matrix that converts C-coordinates into B-coordinates $\left(\underset{C\leftarrow B}{P}\right)^{-1} = \underset{B\leftarrow C}{P}$

Change of Basis in R^n: If $B = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ and E is the standard basis $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ in R^n, then $[\mathbf{b}_1]_E = \mathbf{b}_1$ and likewise for the other vectors in B.
$\underset{E\leftarrow B}{P}$ is the same as the change-of-coordinates matrix P_B (Section 4.4)

To change coordinates between two nonstandard bases in R^n, we need Theorem 15. The theorem shows that to solve the change-of-basis problem, we need the coordinate vectors of the old basis relative to the new basis.

In Ex 2 and 3, let $B = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $C = \{\mathbf{c}_1, \mathbf{c}_2\}$ be bases for R^2. Find the change-of-coordinates matrix from B to C and the change-of-coordinates matrix from C to B. (Hint: For $\underset{C\leftarrow B}{P}$, row reduce $[\mathbf{c}_1\ \mathbf{c}_2\ \mathbf{b}_1\ \mathbf{b}_2]$ and $\underset{B\leftarrow C}{P} = \underset{C\leftarrow B}{P}^{-1}$

Ex 2: $\mathbf{b}_1 = \begin{bmatrix} -1 \\ 8 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}$, $\mathbf{c}_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $\mathbf{c}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$,

Ex 3: $\mathbf{b}_1=\begin{bmatrix}7\\-2\end{bmatrix}$, $\mathbf{b}_2=\begin{bmatrix}2\\-1\end{bmatrix}$, $\mathbf{c}_1=\begin{bmatrix}4\\1\end{bmatrix}$, $\mathbf{c}_2=\begin{bmatrix}5\\2\end{bmatrix}$,

Note: $\begin{bmatrix}c_1 & c_2 \vdots b_1 & b_2\end{bmatrix}\sim\begin{bmatrix}I \vdots \underset{C\leftarrow B}{P}\end{bmatrix}$ or $\begin{bmatrix}P_C & P_B\end{bmatrix}\sim\begin{bmatrix}I & \underset{C\leftarrow B}{P}\end{bmatrix}$

Ex 4: In $\mathbf{P}_2$, find the change-of-coordinates matrix from the basis $B=\{1-3t^2,2+t-5t^2,1+2t\}$ to the standard basis. Then write t^2 as a linear combination of the polynomials in B.

This diagram provides another way of viewing the change-of-coordinates:

$[x]_B \rightarrow P_B[x]B \rightarrow (P_C)^{-1}P_B[x]_B = \underset{C\leftarrow B}{P}[x]_B=[x]_C$

B-coordinates Standard C-coordinates Coordinates

4.9 Applications to Markov Chains

Suppose a physical or mathematical system undergoes a process of change such that at any moment it can be in one of a finite number of states. For example, the weather could be sunny, cloudy, or rainy. Suppose that such a system changes with time from one state to another and at scheduled times the state of the system is observed.

If the probability of a given state can be predicted by just knowing the state of the system at the preceding observation, then the process of change is called a Markov chain or Markov process.

Definition: If a Markov chain has k possible states, which we label as 1, 2, . . . k, then the probability that the system is in state I at any observation after it was in state j at the preceding observation is denoted by p_{ij} and is called the transition probability from state j to state i. The matrix $P = [p_{ij}]$ is called the transition matrix of the Markov chain.

These chains are an expansion of the migration matrices we studied earlier.

For example, in a three state Markov chain, the transition matrix has the form

Preceding State

1	2	3	
p_{11}	p_{12}	p_{13}	1
p_{21}	p_{22}	p_{23}	2 New
p_{31}	p_{32}	p_{33}	3

There is a second important matrix in the use of Markov chains, the state vector.

Definition: The state vector for an observation of a Markov chain with k states is a column vector $\mathbf{x}$ whose i^{th} component x_i is the probability that the system is in the ith state at that time.

Consider the following example: Rent-a-Lemon has three locations from which to rent a car for one day: airport, downtown and the valley

Daily Migration:

	Rented From		
Airport	Downtown	Valley	**Returned to**
.95	.02	.05	Airport
.03	.90	.05	Downtown
.02	.08	.90	Valley

$$M = \begin{bmatrix} .95 & .02 & .05 \\ .03 & .90 & .05 \\ .02 & .08 & .90 \end{bmatrix} \quad \text{(migration matrix)}$$

Note that the probabilities in each column total one. A matrix with this property is called a stochastic matrix.

The initial state vector for this situation is

$$x_0 = \begin{bmatrix} .5 \\ .3 \\ .2 \end{bmatrix} \begin{matrix} \text{(initial fraction of cars at airport)} \\ \text{(initial fraction of cars downtown)} \\ \text{(initial fraction of cars at valley location)} \end{matrix}$$

If we wish to find the distribution of cars after one day, we can find Mx_0

$$Mx_0 = \begin{bmatrix} .95 & .02 & .05 \\ .03 & .90 & .05 \\ .02 & .08 & .90 \end{bmatrix} \begin{bmatrix} .5 \\ .3 \\ .2 \end{bmatrix} =$$

$$x_1 = Mx_0 = \begin{bmatrix} .95 & .02 & .05 \\ .03 & .90 & .05 \\ .02 & .08 & .90 \end{bmatrix} \begin{bmatrix} .5 \\ .3 \\ .2 \end{bmatrix} = \begin{bmatrix} 0.491 \\ 0.295 \\ 0.214 \end{bmatrix}$$

In general,

$$x_{k+1} = Mx_k \ \ for\ k = 0, 1, 2, \ldots$$

So, the distribution after two days =

$$x_2 = Mx_1 = \begin{bmatrix} .95 & .02 & .05 \\ .03 & .90 & .05 \\ .02 & .08 & .90 \end{bmatrix} \begin{bmatrix} 0.491 \\ 0.295 \\ 0.214 \end{bmatrix} = \begin{bmatrix} 0.483 \\ 0.290 \\ 0.226 \end{bmatrix}$$

$$x_3 = Mx_2 = \begin{bmatrix} .95 & .02 & .05 \\ .03 & .90 & .05 \\ .02 & .08 & .90 \end{bmatrix} \begin{bmatrix} 0.483 \\ 0.290 \\ 0.226 \end{bmatrix} = \begin{bmatrix} 0.475 \\ 0.287 \\ 0.236 \end{bmatrix}$$

$$x_4 = Mx_3 = \begin{bmatrix} .95 & .02 & .05 \\ .03 & .90 & .05 \\ .02 & .08 & .90 \end{bmatrix} \begin{bmatrix} 0.475 \\ 0.287 \\ 0.236 \end{bmatrix} = \begin{bmatrix} 0.468 \\ 0.284 \\ 0.244 \end{bmatrix}$$

$\vdots$

$$x_{49} = Mx_{48} = \begin{bmatrix} 0.417 \\ 0.278 \\ 0.305 \end{bmatrix}$$

$$x_{50} = Mx_{49} = \begin{bmatrix} 0.417 \\ 0.278 \\ 0.305 \end{bmatrix} \text{ (long term distribution)}$$

-
-
-

$\mathbf{x} = \begin{bmatrix} 0.417 \\ 0.278 \\ 0.305 \end{bmatrix}$ is called a **steady state vector**

since $\mathbf{x} = M\mathbf{x}$

Ex 1: By reviewing its donation records, the alumni office of a college finds that 80% of its alumni who contribute to the annual fund one year will also contribute the next year, and 30% of those who do not contribute one year will contribute the next. Construct a transition matrix and initial state matrix illustrating this situation. Find the contributions for each of the first five years. Continue until you find the steady state vector.

There is process that can be used to find the steady state vector so that it is not necessary to use trial and error.

Finding the Steady State Vector

We know that in the long run

$$M\mathbf{x} = \mathbf{x}$$

So,

$$M\mathbf{x} = I\mathbf{x}$$

Subtracting, we get,

$$M\mathbf{x} - I\mathbf{x} = \mathbf{0}$$

Factoring, we get

$$(M - I)\mathbf{x} = \mathbf{0}$$

Solve $(M - I)\mathbf{x} = \mathbf{0}$ to find the steady state vector.

Note that the solution $\mathbf{x}$ must be a probability vector.

Ex 2: Find the steady state vectors for each of the following transition matrices

a. $\begin{bmatrix} .81 & .26 \\ .19 & .74 \end{bmatrix}$

b. $\begin{bmatrix} \frac{1}{3} & \frac{1}{2} & 0 \\ \frac{1}{3} & 0 & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{2} & \frac{3}{4} \end{bmatrix}$

Ex 3: Suppose that 3% of the population of the U.S. lives in the State of Washington. Suppose the migration of the population into and out of Washington State will be constant for many years according to the following migration probabilities. What percentage of the total U.S. population will eventually live in Washington?

From

WA	Rest of U.S.	To:
0.9	0.01	WA
0.10	0.99	Rest of U. S.